# Probability Rules & Conditional Probability

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Basic probability rules**

A fair six-sided die is rolled. Find the probability the roll is even, and the probability the roll is greater than 4.

$P(\text{even}) = P(2, 4, 6) = \frac{3}{6} = 0.5$. $P(\text{greater than } 4) = P(5, 6) = \frac{2}{6} \approx 0.333$. (Written in words: the chance of rolling even is $0.5$; the chance of rolling above $4$ is about $0.333$.)

**2** **[Foundational]** • **Target: Basic probability rules**

For events $A$ and $B$, $P(A) = 0.4$, $P(B) = 0.3$, and $P(A \cup B) = 0.6$. Find $P(A \cap B)$ using the addition rule.

Addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. So $0.6 = 0.4 + 0.3 - P(A \cap B)$, giving $P(A \cap B) = 0.7 - 0.6 = 0.1$.

**3** **[Foundational]** • **Target: Mutually exclusive and independent events**

Explain the difference between mutually exclusive events and independent events, and give a brief example of each.

Mutually exclusive events cannot happen at the same time (e.g. rolling a 2 and rolling a 5 on the same single die roll — $P(A \cap B) = 0$). Independent events do not affect each other's probability of occurring (e.g. flipping heads on one coin and rolling a 6 on a separate die — knowing one outcome tells you nothing about the other).

**4** **[Foundational]** • **Target: Mutually exclusive and independent events**

Two events $A$ and $B$ satisfy $P(A) = 0.5$ and $P(B) = 0.5$, and they are mutually exclusive. Explain why $A$ and $B$ cannot also be independent (unless one has probability $0$).

If $A$ and $B$ are independent, $P(A \cap B) = P(A)P(B) = 0.25$. But since they are mutually exclusive, $P(A \cap B) = 0$. Since $0.25 \neq 0$, the two events cannot be both mutually exclusive and independent here — in general, two events with positive probability cannot be both mutually exclusive and independent, since mutual exclusivity means one occurring rules out the other, which is a strong dependence, not independence.

**5** **[Foundational]** • **Target: Conditional probability**

In a class, the probability a student likes math is $0.6$, and the probability a student likes both math and science is $0.42$. Find the probability a student likes science given that they like math.

$P(\text{science given math}) = \frac{P(\text{math and science})}{P(\text{math})} = \frac{0.42}{0.6} = 0.7$.

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**6** [Foundational] • Target: Conditional probability

Use the multiplication rule to find $P(A \cap B)$ if $P(A) = 0.5$ and $P(B \mid A) = 0.4$.

Multiplication rule: $P(A \cap B) = P(A) \times P(B \mid A) = 0.5 \times 0.4 = 0.2$.

**7** [Foundational] • Target: Two-way tables and conditional probability

A survey of 200 students finds 90 own a car, 120 have a part-time job, and 60 both own a car and have a job. Find the probability a randomly selected student has a job, given that they own a car.

$P(\text{job given car}) = \frac{P(\text{car and job})}{P(\text{car})} = \frac{60/200}{90/200} = \frac{60}{90} = \frac{2}{3} \approx 0.667$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Two-way tables and conditional probability

Using the same survey data, determine whether owning a car and having a job are independent events, showing your reasoning.

$P(\text{job}) = \frac{120}{200} = 0.6$. From the previous problem, $P(\text{job given car}) \approx 0.667$. Since these two probabilities are not equal ($0.667 \neq 0.6$), owning a car and having a job are not independent — knowing a student owns a car changes the probability they have a job.

**9** [Intermediate] • Target: Tree diagrams and sequential events

A factory has two machines: Machine A produces 60% of items with a 2% defect rate, and Machine B produces 40% of items with a 5% defect rate. Find the probability a randomly selected item is defective, using the law of total probability.

$P(\text{defective}) = P(A)P(\text{defective} \mid A) + P(B)P(\text{defective} \mid B) = (0.6)(0.02) + (0.4)(0.05) = 0.012 + 0.02 = 0.032$, or $3.2\%$.

**10** [Intermediate] • Target: Applying probability rules to real contexts

A weather forecaster states there is a 30% chance of rain on Saturday and a 30% chance of rain on Sunday, and treats the two days as independent. Find the probability it rains on both days, and the probability it rains on at least one of the two days.

Both days: $P(\text{Sat and Sun}) = 0.3 \times 0.3 = 0.09$. At least one day: using the complement, $P(\text{neither day}) = (1 - 0.3)(1 - 0.3) = 0.7 \times 0.7 = 0.49$, so $P(\text{at least one}) = 1 - 0.49 = 0.51$.

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**11** [Intermediate] • Target: The complement rule

A box of 50 light bulbs contains 6 defective bulbs. If one bulb is selected at random, find the probability it is NOT defective, using the complement rule.

$P(\text{defective}) = \frac{6}{50} = 0.12$. $P(\text{not defective}) = 1 - 0.12 = 0.88$.

**12** [Intermediate] • Target: The complement rule

A student estimates there is a 65% chance of rain at least once during a 3-day trip, assuming each day is independent with the same daily rain probability $p$. Find $p$, to three decimal places, using $P(\text{no rain all 3 days}) = (1-p)^3$.

$P(\text{at least one rainy day}) = 1 - (1-p)^3 = 0.65$, so $(1-p)^3 = 0.35$. Taking the cube root: $1 - p = (0.35)^{1/3} \approx 0.705$, giving $p \approx 0.295$.

**13** [Intermediate] • Target: Combining independence and conditional probability

Two lie detector tests are given to a suspect, each correctly detecting deception 90% of the time (independently of each other). If the suspect is actually lying, find the probability both tests correctly detect it.

Since the tests are independent: $P(\text{both correct}) = 0.9 \times 0.9 = 0.81$.

**14** [Intermediate] • Target: Combining independence and conditional probability

A bag contains 5 red marbles and 3 blue marbles. Two marbles are drawn without replacement. Find the probability both are red.

$P(\text{first red}) = \frac{5}{8}$. Given the first is red, $P(\text{second red} \mid \text{first red}) = \frac{4}{7}$.
$P(\text{both red}) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14} \approx 0.357$.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Combining independence and conditional probability • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

At a certain hospital, 40% of patients admitted have condition A, and 25% have condition B. Additionally, 10% of all patients have both conditions A and B. Draw on your understanding of the addition rule to find the probability a randomly selected patient has condition A or condition B (or both), and separately explain why simply adding $P(A) + P(B)$ without adjustment would overcount certain patients.

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.40 + 0.25 - 0.10 = 0.55$. Simply adding $P(A) + P(B) = 0.65$ would double-count the patients who have both conditions (the 10% overlap), since those patients would be counted once within the 40% and again within the 25% — subtracting $P(A \cap B)$ corrects for this double-counting.

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**16** **[Challenging]** • **Target: Combining independence and conditional probability** • **Cross-Lesson: 1.01 Measures of Center & Spread**

An airline reports that 85% of its flights depart on time. Among on-time flights, 95% also arrive on time, while among delayed departures, only 40% arrive on time. Using a tree-diagram approach and the law of total probability, find the overall probability that a randomly selected flight arrives on time.

$P(\text{arrives on time}) = P(\text{on – time departure}) \times P(\text{on – time arrival} \mid \text{on – time departure}) + P(\text{delayed depa}$ or about $86.75\%$.

**17** **[Challenging]** • **Target: Combining independence and conditional probability** • **Cross-Lesson: 1.02 Z-Scores & Linear Transformations**

A quality inspector at a factory knows that 3% of all items produced are defective. The inspection process correctly identifies a defective item 98% of the time, but also incorrectly flags a non-defective item as defective 2% of the time (a false positive). If an item is flagged as defective by the inspection process, use the given information and the multiplication rule to find the probability that the item actually IS defective, showing every step of your reasoning (this requires combining the law of total probability with conditional probability, similar to Bayes' theorem).

$P(\text{defective and flagged}) = P(\text{defective}) \times P(\text{flagged} \mid \text{defective}) = (0.03)(0.98) = 0.0294$.
$P(\text{not defective and flagged}) = P(\text{not defective}) \times P(\text{flagged} \mid \text{not defective}) = (0.97)(0.02) = 0.0194$
. $P(\text{flagged}) = 0.0294 + 0.0194 = 0.0488$.
$P(\text{defective} \mid \text{flagged}) = \frac{P(\text{defective and flagged})}{P(\text{flagged})} = \frac{0.0294}{0.0488} \approx 0.603$, or about $60.3\%$ — notably less than the 98% accuracy rate might naively suggest, because defects are rare to begin with.

**18** **[Challenging]** • **Target: Combining independence and conditional probability** • **Cross-Lesson: 1.03 Describing Distributions & Linear Regression**

A survey of 500 adults finds that 280 exercise regularly, 220 eat a balanced diet, and 150 do both. Using this information, first find the probability a randomly selected adult exercises regularly or eats a balanced diet (or both), and then find the probability an adult eats a balanced diet given that they exercise regularly, showing every step of your work clearly.

$P(\text{exercise}) = \frac{280}{500} = 0.56$. $P(\text{diet}) = \frac{220}{500} = 0.44$. $P(\text{both}) = \frac{150}{500} = 0.30$. Using the addition rule:
$P(\text{exercise or diet}) = 0.56 + 0.44 - 0.30 = 0.70$. Using the conditional probability formula:
$P(\text{diet} \mid \text{exercise}) = \frac{P(\text{both})}{P(\text{exercise})} = \frac{0.30}{0.56} \approx 0.536$, or about $53.6\%$.

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**19** **[Challenging]** • **Target: The complement rule** • **Cross-Lesson: 2.01 Sampling Methods & Bias**

Events $A$ and $B$ satisfy $P(A) = 0.40$, $P(B) = 0.50$, and $P(A \cup B) = 0.70$. (a) Find $P(A \cap B)$. (b) Determine whether $A$ and $B$ are independent. (c) Determine whether $A$ and $B$ are mutually exclusive.

(a) Intersection: $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.40 + 0.50 - 0.70 = \mathbf{0.20}$. (b) Independence: Test whether $P(A \cap B) = P(A)P(B)$: $P(A)P(B) = (0.40)(0.50) = 0.20$. Because $P(A \cap B) = P(A)P(B) = 0.20$, events $A$ and $B$ are independent. (c) Mutually Exclusive: Because $P(A \cap B) = 0.20 eq 0$, events $A$ and $B$ can occur simultaneously and are therefore not mutually exclusive.

**20** **[Challenging]** • **Target: Combining independence and conditional probability** • **Cross-Lesson: 2.02 Experimental Design & Scope of Inference**

A national health organization is investigating the relationship between daily screen time (in hours) and the number of hours of sleep reported by adults. They collect data from a sample of 1200 adults. The data is summarized in a two-way table, where the row totals represent categories of daily screen time and the column totals represent categories of reported sleep hours. The organization is particularly interested in the cells where adults report less than 4 hours of screen time and get 7 or more hours of sleep. If the overall row total for 'less than 4 hours of screen time' is 250, and the overall column total for '7 or more hours of sleep' is 300, what is the expected count for the cell representing adults with less than 4 hours of screen time who get 7 or more hours of sleep, assuming independence between screen time and sleep hours?

The problem asks for the expected count in a specific cell of a contingency table, assuming independence between the two categorical variables: daily screen time and reported sleep hours. The formula for the expected count in a cell is given by: $E = \frac{\text{(Row Total)} \times \text{(Column Total)}}{\text{Grand Total}}$ In this scenario: - The Grand Total sample size is given as 1200 adults. - The Row Total for the category 'less than 4 hours of screen time' is 250. - The Column Total for the category '7 or more hours of sleep' is 300. We want to find the expected count for the cell where both these conditions are met (less than 4 hours of screen time AND 7 or more hours of sleep). Applying the formula: $E = \frac{250 \times 300}{1200}$ $E = \frac{75000}{1200}$ $E = 62.50$ Therefore, the expected count for this cell is 62.50. This represents the count we would expect in this cell if there were no association between daily screen time and reported sleep hours in the population from which the sample was drawn. A common misconception might be to use only one of the totals or to misapply the formula, leading to incorrect answers.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: Basic probability rules

For events $A$ and $B$, $P(A) = 0.30$, $P(B) = 0.70$, and $P(A \cap B) = 0.05$. Find $P(A \cup B)$ using the addition rule.

**a.** 0.90

**b.** 1.05

**c.** 0.95 ✓

**d.** 1.00

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.30 + 0.70 - 0.05 = 0.95$.

**22** [Foundational] • Target: Mutually exclusive and independent events

For events $A$ and $B$, $P(A) = 0.35$, $P(B) = 0.60$, and $P(A \cap B) = 0.25$. Find $P(A \cup B)$ using the addition rule.

**a.** 0.70 ✓

**b.** 0.95

**c.** 0.75

**d.** 0.65

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.35 + 0.60 - 0.25 = 0.70$.

**23** [Foundational] • Target: Conditional probability

For events $A$ and $B$, $P(A) = 0.25$, $P(B) = 0.55$, and $P(A \cap B) = 0.20$. Find $P(A \cup B)$ using the addition rule.

**a.** 0.55

**b.** 0.80

**c.** 0.65

**d.** 0.60 ✓

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.25 + 0.55 - 0.20 = 0.60$.

**24** [Foundational] • Target: Two-way tables and conditional probability

For events $A$ and $B$, $P(A) = 0.35$, $P(B) = 0.50$, and $P(A \cap B) = 0.10$. Find $P(A \cup B)$ using the addition rule.

**a.** 0.75 ✓

**b.** 0.80

**c.** 0.70

**d.** 0.85

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.35 + 0.50 - 0.10 = 0.75$.

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**25** [Foundational] • Target: Tree diagrams and sequential events

For events $A$ and $B$, $P(A) = 0.35$, $P(B) = 0.55$, and $P(A \cap B) = 0.30$. Find $P(A \cup B)$ using the addition rule.

**a.** 0.65

**b.** 0.90

**c.** 0.55

**d.** 0.60 ✓

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.35 + 0.55 - 0.30 = 0.60.$

**26** [Foundational] • Target: Applying probability rules to real contexts

For events $A$ and $B$, $P(A) = 0.60$, $P(B) = 0.65$, and $P(A \cap B) = 0.25$. Find $P(A \cup B)$ using the addition rule.

**a.** 0.95

**b.** 1.00 ✓

**c.** 1.05

**d.** 1.25

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.60 + 0.65 - 0.25 = 1.00.$

**27** [Foundational] • Target: The complement rule

For events $A$ and $B$, $P(A) = 0.55$, $P(B) = 0.45$, and $P(A \cap B) = 0.45$. Find $P(A \cup B)$ using the addition rule.

**a.** 0.50

**b.** 0.55 ✓

**c.** 0.60

**d.** 1.00

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.55 + 0.45 - 0.45 = 0.55.$

**28** [Foundational] • Target: Combining independence and conditional probability

For events $A$ and $B$, $P(A) = 0.30$ and $P(A \cap B) = 0.30$. Find $P(B \mid A)$.

**a.** 1.10

**b.** 1.00 ✓

**c.** 0.90

**d.** 0.00

$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{0.30}{0.30} \approx 1.00.$

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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**29** [Intermediate] • Target: General Addition Rule & Mutually Exclusive Events

For events $A$ and $B$, $P(A) = 0.30$ and $P(A \cap B) = 0.10$. Find $P(B \mid A)$.

**a.** 0.43

**b.** 0.20

**c.** 0.33 ✓

**d.** 0.23

$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{0.10}{0.30} \approx 0.33.$

**30** [Intermediate] • Target: Conditional Probability Formula

For events $A$ and $B$, $P(A) = 0.80$ and $P(A \cap B) = 0.40$. Find $P(B \mid A)$.

**a.** 0.60

**b.** 0.50 ✓

**c.** 0.40

**d.** 0.55

$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{0.40}{0.80} \approx 0.50.$

**31** [Intermediate] • Target: Multiplication Rule and Tree Diagrams

For events $A$ and $B$, $P(A) = 0.65$ and $P(A \cap B) = 0.60$. Find $P(B \mid A)$.

**a.** 0.92 ✓

**b.** 1.02

**c.** 0.82

**d.** 0.05

$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{0.60}{0.65} \approx 0.92.$

**32** [Intermediate] • Target: Bayes' Theorem in Context

For events $A$ and $B$, $P(A) = 0.70$ and $P(A \cap B) = 0.60$. Find $P(B \mid A)$.

**a.** 0.10

**b.** 0.86 ✓

**c.** 0.76

**d.** 0.96

$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{0.60}{0.70} \approx 0.86.$

**33** [Intermediate] • Target: Basic probability rules

For events $A$ and $B$, $P(A) = 0.70$ and $P(A \cap B) = 0.35$. Find $P(B \mid A)$.

**a.** 0.50 ✓

**b.** 0.35

**c.** 0.60

**d.** 0.40

$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{0.35}{0.70} \approx 0.50.$

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**34** [Intermediate] • Target: Mutually exclusive and independent events

For events $A$ and $B$, $P(A) = 0.35$ and $P(A \cap B) = 0.10$. Find $P(B \mid A)$.

**a.** $0.29$ ✓

**b.** $0.39$

**c.** $0.25$

**d.** $0.19$

$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{0.10}{0.35} \approx 0.29$.

**35** [Intermediate] • Target: Conditional probability

A bag contains $7$ red marbles and $6$ blue marbles. Two marbles are drawn without replacement. Find the probability both are blue.

**a.** $\frac{19}{65}$

**b.** $\frac{6}{65}$

**c.** $\frac{6}{13}$

**d.** $\frac{5}{26}$ ✓

P(first blue) $= \frac{6}{13}$. Given the first is blue, P(second blue given first blue) $= \frac{5}{12}$. P(both blue) $= \frac{6}{13} \times \frac{5}{12} = \frac{5}{26}$.

**36** [Intermediate] • Target: Two-way tables and conditional probability

A bag contains $4$ red marbles and $8$ blue marbles. Two marbles are drawn without replacement. Find the probability both are blue.

**a.** $\frac{107}{330}$

**b.** $\frac{14}{33}$ ✓

**c.** $\frac{2}{3}$

**d.** $\frac{173}{330}$

P(first blue) $= \frac{8}{12}$. Given the first is blue, P(second blue given first blue) $= \frac{7}{11}$. P(both blue) $= \frac{8}{12} \times \frac{7}{11} = \frac{14}{33}$.

**37** [Intermediate] • Target: Tree diagrams and sequential events

A factory has two machines: Machine A produces $80\%$ of items with a $10\%$ defect rate, and Machine B produces $20\%$ of items with a $3\%$ defect rate. Letting $D$ denote the event that an item is defective, find $P(D)$ for a randomly selected item, using the law of total probability.

**a.** $8.60\%$ ✓

**b.** $4.40\%$

**c.** $8.10\%$

**d.** $9.10\%$

$P(D) = P(A)P(D \mid A) + P(B)P(D \mid B) = (0.80)(0.10) + (0.20)(0.03) = 8.60\%$.

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** **[Challenging]** • **Target: Applying probability rules to real contexts** • **Cross-Lesson: 1.03 Describing Distributions & Linear Regression**

A factory has two machines: Machine A produces 50% of items with a 9% defect rate, and Machine B produces 50% of items with a 2% defect rate. Letting $D$ denote the event that an item is defective, find $P(D)$ for a randomly selected item, using the law of total probability.

**a.** 6.00%

**b.** 5.50% ✓

**c.** 6.50%

**d.** 5.00%

$P(D) = P(A)P(D \mid A) + P(B)P(D \mid B) = (0.50)(0.09) + (0.50)(0.02) = 5.50\%.$

**39** **[Challenging]** • **Target: The complement rule** • **Cross-Lesson: 2.01 Sampling Methods & Bias**

A quality inspector knows 5% of all items produced are defective (event $D$). The inspection process correctly flags a defective item 93% of the time, but also flags a non-defective item as defective 6% of the time (a false positive). If an item is flagged as defective (event $F$), find $P(D \mid F)$, to one decimal place.

**a.** 93.0%

**b.** 47.9%

**c.** 44.9% ✓

**d.** 41.9%

$P(D \cap F) = P(D)P(F \mid D) = (0.05)(0.93) = 0.0465.$
$P(D^c \cap F) = P(D^c)P(F \mid D^c) = (0.95)(0.06) = 0.0570.$ $P(F) = 0.1035.$ $P(D \mid F) = \frac{0.0465}{0.1035} \approx 44.9\%.$

**40** **[Challenging]** • **Target: Combining independence and conditional probability** • **Cross-Lesson: 2.02 Experimental Design & Scope of Inference**

A quality inspector knows 3% of all items produced are defective (event $D$). The inspection process correctly flags a defective item 97% of the time, but also flags a non-defective item as defective 6% of the time (a false positive). If an item is flagged as defective (event $F$), find $P(D \mid F)$, to one decimal place.

**a.** 30.3%

**b.** 97.0%

**c.** 33.3% ✓

**d.** 36.3%

$P(D \cap F) = P(D)P(F \mid D) = (0.03)(0.97) = 0.0291.$
$P(D^c \cap F) = P(D^c)P(F \mid D^c) = (0.97)(0.06) = 0.0582.$ $P(F) = 0.0873.$ $P(D \mid F) = \frac{0.0291}{0.0873} \approx 33.3\%.$

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**41** [Challenging] • Target: General Addition Rule & Mutually Exclusive Events • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

If $P(A) = 0.40$, $P(B) = 0.35$, and $P(A \cap B) = 0.15$, what is $P(A \cup B)$?

**a.** $0.60$ ✓

**b.** $0.75$

**c.** $0.90$

**d.** $0.25$

By the General Addition Rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.40 + 0.35 - 0.15 = 0.60$.

**42** [Challenging] • Target: General Addition Rule & Mutually Exclusive Events • Cross-Lesson: 3.03 The Normal Distribution

Can two events with non-zero probabilities be both mutually exclusive and independent?

**a.** **No, if they are mutually exclusive, the occurrence of one prevents the other, meaning they are dependent** ✓

**b.** Yes, mutually exclusive and independent are synonyms in probability

**c.** Only if the sum of their probabilities equals 1

**d.** Yes, as long as their intersection is zero

If $A$ and $B$ are disjoint, $P(A \cap B) = 0$. For independence, $P(A \cap B) = P(A)P(B) > 0$. Since $0 \neq P(A)P(B)$, they cannot be both mutually exclusive and independent.

**43** [Challenging] • Target: Conditional Probability Formula • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

Which statement mathematically proves that two events $A$ and $B$ are independent?

**a.** $P(A \mid B) = P(A)$ ✓

**b.** $P(A \cup B) = P(A) + P(B)$

**c.** $P(A \cap B) = 0$

**d.** $P(A) = P(B)$

Events $A$ and $B$ are independent if knowing that $B$ occurred does not change the probability of $A$, meaning $P(A \mid B) = P(A)$ (or $P(A \cap B) = P(A)P(B)$).

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**44** [Challenging] • Target: Multiplication Rule and Tree Diagrams • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

If two events $C$ and $D$ are independent with $P(C) = 0.60$ and $P(D) = 0.70$, what is $P(C \cap D)$?

**a.** $0.42$ ✓

**b.** $1.30$

**c.** $0.10$

**d.** $0.65$

For independent events, $P(C \cap D) = P(C) \times P(D) = 0.60 \times 0.70 = 0.42$.

**45** [Challenging] • Target: Bayes' Theorem in Context • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

Continuing the factory problem: if an item is found to be defective, what is the probability it came from Machine A?

**a.** $\frac{0.012}{0.032} = 0.375$ ✓

**b.** $\frac{0.60}{0.032}$

**c.** $0.60$

**d.** $0.020$

By Bayes' Theorem: $P(A \mid D) = \frac{P(A \cap D)}{P(D)} = \frac{0.012}{0.032} = \frac{3}{8} = 0.375$.

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